

Q1a) What is Model Building? Explain this phase of Data Analytics with suitable example.

Model Building is an important phase in the Data Analytics life cycle in which mathematical, statistical, or machine learning models are created using prepared data to solve business problems and make predictions or decisions.

The objective of model building is to find patterns and relationships in data and generate accurate results.

Steps in Model Building Phase

1. Select Modeling Technique

Choose an appropriate technique depending on the problem: Classification, Regression, Clustering, Decision Trees, Neural Networks etc.

2. Split Data

Dataset is divided into:

- Training Data → used to train model
- Testing Data → used to evaluate model

3. Build the Model

Apply algorithms on training data.

4. Validate the Model

Check model accuracy using Accuracy, Precision, Recall, RMSE etc.

5. Tune Parameters

Improve performance by adjusting parameters.

6. Finalize Model

Best performing model is selected for deployment.

Suitable Example

Example: Student Result Prediction System

Suppose a college wants to predict whether a student will pass or fail.

Step 1: Collect Data

Data includes Attendance, Internal marks, Assignment submission, Study hours

Step 2: Choose Model

Use Classification Algorithm like Decision Tree.

Step 3: Train Model

Historical student data is used to train the model.

Step 4: Test Model

New student data is tested.

Step 5: Prediction

Model predicts:

- Pass
- Fail

Outcome - College can identify weak students early and provide guidance.

Conclusion

Model building transforms raw data into useful predictive systems. It helps organizations make accurate and data-driven decisions.

Q1b) Different Stakeholders of an Analytics Project and Their Expected Outputs

Stakeholders in Analytics Project

Stakeholders are individuals or groups involved in or affected by the analytics project. For a successful analytics project, different professionals work together with specific responsibilities.

1. Business Users

- Use analytics results for decision-making.
- Expect actionable insights.

2. Project Sponsor

- Provides funding and business support.
- Defines business objectives.

3. Project Manager

- Manages schedule, resources, and communication.
- Completes project within time and budget.

4. Business Analyst

- Understands business requirements.
- Connects technical and business teams.

5. Database Administrator (DBA)

- Maintains databases, Ensures security and efficient storage

6. Data Engineers

- Handle data collection, storage, and preprocessing.

7. Data Scientists

- Build analytical and predictive models.

8. IT Team

- Provides infrastructure and deployment support.

9. End Users/Customers

- Final users benefiting from the system.

Expected Outputs at Conclusion of Project

Stakeholder	Expected Output
-------------	-----------------

Project Sponsor	Business value, ROI, profit growth
-----------------	------------------------------------

Business Users	Reports, dashboards, predictions
----------------	----------------------------------

Project Manager	Successful completion within time and budget
-----------------	--

Data Scientists	Accurate and validated models
-----------------	-------------------------------

Data Engineers	Clean and reliable data pipelines
----------------	-----------------------------------

IT Team	Stable deployment and maintenance
---------	-----------------------------------

End Users	Better services and user experience
-----------	-------------------------------------

Key Deliverables of Analytics Project - Analytical reports, Predictive models, Dashboards and visualizations, Business recommendations, Documentation, Performance metrics.

Conclusion: Different stakeholders have different expectations from an analytics project. Successful analytics projects provide technical solutions as well as business value to all stakeholders.

Q2a) Explain Descriptive, Diagnostic and Predictive Analytics

Analytics is the process of analyzing data to gain useful insights and support decision-making.

1. Descriptive Analytics

Descriptive Analytics explains **what happened in the past** by analyzing historical data. It summarizes data using Report, Charts, Dashboards, Graphs.

Features

- Uses historical data
- Gives summary of events
- Helps understand business performance

Examples

- Monthly sales report
- Student result analysis
- Website traffic report

Techniques Used: Data aggregation, Data mining, Reporting, Visualization.

Example: An e-commerce company analyzes last year's sales data to know:

- Best-selling products
- Total revenue
- Customer growth

2. Diagnostic Analytics

Diagnostic Analytics explains **why something happened**.

It identifies causes and relationships behind a problem or event.

Features

- Finds root cause of problems
- Uses drill-down and correlation analysis
- Helps in problem solving

Techniques Used: Data discovery, Correlation analysis, Querying, Statistical analysis

Examples

- Why sales decreased?
- Why students failed?
- Why website traffic dropped?

Example

A company finds that sales dropped because:

- Product price increased
- Competitor launched new product
- Marketing campaign failed

3. Predictive Analytics

Predictive Analytics predicts **future outcomes** using historical and current data.

It uses statistical and machine learning models.

Features

- Forecasts future trends
- Supports decision-making
- Uses probability and prediction

Techniques Used: Regression, Decision Trees, Machine Learning, Neural Networks

Examples

- Weather forecasting
- Stock market prediction
- Predicting customer behavior

Example

An online shopping company predicts:

- Which customers may buy products
- Future sales trends
- Customer churn

Comparison Table

Type	Purpose	Question Answered	Example
Descriptive Analytics	Understand past	What happened?	Sales report
Diagnostic Analytics	Find causes	Why did it happen?	Reason for sales drop
Predictive Analytics	Predict future	What may happen?	Future sales prediction

Conclusion

Descriptive analytics summarizes past events, diagnostic analytics identifies causes, and predictive analytics forecasts future outcomes. Together they help organizations make better decisions.

Q2b) Activities involved in Identifying Potential Data Resources during Discovery Phase of Data Analytics Life Cycle

Discovery Phase

The Discovery Phase is the first phase of the Data Analytics Life Cycle. In this phase, analysts understand the business problem, objectives, available resources, and possible data sources.

Identifying potential data resources is one of the major activities in this phase.

Activities Involved in Identifying Potential Data Resources

1. Understanding Business Problem

- Understand project goals and requirements.
- Identify what type of data is needed.

Example

For sales prediction Customer data, Product data, Sales records are required.

2. Identifying Internal Data Sources

Data available inside the organization is identified.

Sources: Databases, ERP systems, CRM systems, Transaction records, Server logs

Example

Company sales database.

3. Identifying External Data Sources

Data collected from outside the organization.

Sources: Social media, Government datasets, Websites, APIs, Market surveys

Example

Twitter customer reviews.

4. Data Collection Planning

Plan how data will be collected.

Activities: Define methods, Decide tools, Set collection frequency

5. Data Acquisition

Acquire data from identified sources.

Methods: Import files, Database queries, APIs, Web scraping

6. Data Quality Assessment

Check whether data is Complete, Accurate, Consistent, Reliable

Example

Removing duplicate or missing records.

7. Data Integration

Combine data from multiple sources into one format.

Example

Merging customer and sales data.

8. Evaluate Data Relevance

Verify whether collected data supports business objectives.

Example

Removing unnecessary attributes.

Conclusion

Identifying potential data resources is a critical activity in the Discovery Phase because high-quality and relevant data leads to accurate analytics and better business decisions.

Q1a) Draw Data Analytics Life Cycle Diagram and Explain its Phases

Data Analytics Life Cycle

The Data Analytics Life Cycle is a systematic process followed to solve business problems using data analysis techniques.

Diagram of Data Analytics Life Cycle

Phases of Data Analytics Life Cycle -

1. Discovery Phase

Purpose: Understand the business problem and project objectives.

Activities: Identify business goals, Collect initial requirements, Identify data sources, Understand available resources

Output: Project plan and business understanding.

2. Data Preparation Phase

Purpose: Prepare data for analysis.

Activities: Data collection, Data cleaning, Data transformation, Handling missing values

Output: Clean and structured dataset.

3. Model Planning Phase

Purpose: Select suitable analytical techniques and models.

Activities: Choose algorithms, Select tools and methods, Define training/testing strategy

Output: Model design and planning.

4. Model Building Phase

Purpose: Build and test analytical models.

Activities: Train models, Validate accuracy, Tune parameters

Output: Final predictive or analytical model.

5. Communicate Results Phase

Purpose: Present findings to stakeholders.

Activities: Prepare reports, Create visualizations, Explain insights and recommendations

Output: Business insights and reports.

6. Operationalize Phase

Purpose: Deploy model into real environment.

Activities: Deploy system, Monitor performance, Maintain model

Output: Working analytics solution.

Conclusion

The Data Analytics Life Cycle provides a structured approach for converting raw data into meaningful insights and business solutions.

Q1c) What are Various Sources of Big Data?

Big Data is generated from various structured, semi-structured, and unstructured sources at very high speed and volume.

Various Sources of Big Data

1. Social Media Data

Generated from: Facebook, Twitter, Instagram, YouTube

Examples: Posts, comments, likes, videos.

2. Machine Generated Data

Generated automatically by machines and devices.

Examples: Sensor data, IoT devices, RFID tags, Smart meters

3. Transactional Data

Generated during business transactions.

Examples: Online shopping, Banking transactions, ATM transactions

4. Web and Clickstream Data

Generated from websites and online activities.

Examples: Search history, Page visits, User clicks

5. Mobile Data

Generated from smartphones and mobile applications.

Examples: GPS location, App usage, Call records

6. Scientific Data

Generated from scientific research and experiments.

Examples: Satellite images, Weather forecasting, Medical research

7. Enterprise Data

Generated inside organizations.

Examples: Emails, Employee records, ERP systems, CRM systems

8. Multimedia Data

Includes audio, video, and image files.

Examples: CCTV footage, Photos, Streaming videos

Conclusion

Big Data comes from multiple diverse sources such as social media, IoT devices, business transactions, and multimedia platforms, helping organizations gain valuable insights.

Q2a) Describe Few Applications of Big Data Analytics

Big Data Analytics is the process of analyzing large and complex datasets to discover useful patterns, trends, and insights for better decision-making.

Applications of Big Data Analytics

1. Healthcare

Uses - Disease prediction, Patient monitoring, Medical research

Example - Hospitals analyze patient records to predict diseases and improve treatment.

2. Banking and Finance

Uses - Fraud detection, Risk analysis, Customer analytics

Example - Banks detect suspicious transactions using analytics.

3. E-Commerce

Uses - Product recommendation, Customer behavior analysis, Sales prediction

Example - Online shopping websites recommend products based on customer interests.

4. Social Media Analytics

Uses - Sentiment analysis, User behavior tracking, Advertisement targeting

Example - Companies analyze tweets and comments to know customer opinions.

5. Transportation

Uses - Traffic prediction, Route optimization, Vehicle tracking

Example - Google Maps predicts traffic using real-time data.

6. Education

Uses - Student performance analysis, Personalized learning, Attendance monitoring

Example - Colleges identify weak students using analytics.

7. Agriculture

Uses - Crop monitoring, Weather analysis, Soil analysis

Example - Farmers use sensor data to improve crop production.

8. Manufacturing

Uses - Predictive maintenance, Quality control, Production optimization

Example - Factories use sensors to detect machine failures early.

Conclusion

Big Data Analytics is widely used in healthcare, banking, education, transportation, and many other fields to improve efficiency and decision-making.

Q2b) Explain Data Preparation Phase of Data Analytics Lifecycle

Data Preparation is the second phase of the Data Analytics Life Cycle in which raw data is collected, cleaned, transformed, and organized for analysis.

It is one of the most important phases because the quality of analysis depends on the quality of data.

Activities in Data Preparation Phase

1. Data Collection

Collect data from different sources such as Databases, Websites, Sensors, Social media.

2. Data Cleaning

Remove errors and inconsistencies from data.

Activities: Handle missing values, Remove duplicate records, Correct incorrect data

3. Data Integration

Combine data from multiple sources into a single dataset.

Example: Merging customer data with sales data.

4. Data Transformation

Convert data into suitable format for analysis.

Activities: Normalization, Aggregation, Encoding

5. Data Reduction

Reduce data size while preserving important information.

Methods - Sampling, Feature selection

6. Data Formatting

Organize data into proper structure such as Tables, CSV files, Dataframes

Importance of Data Preparation

- Improves data quality
- Increases model accuracy
- Reduces processing time
- Helps better decision-making

Conclusion

The data preparation phase converts raw data into clean and usable data, which is essential for accurate analytics and model building.

Analytics Sandbox is a secure and isolated environment used by data analysts and data scientists to explore, process, and analyze data without affecting the original production systems.

It acts as a workspace for experimentation and model development.

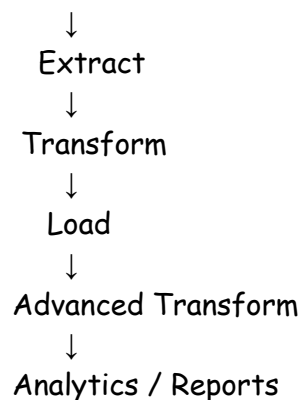
Features of Analytics Sandbox - Stores large volumes of data, Supports data exploration, Allows testing of models, Does not affect live systems, Provides high processing capability

Advantages - Safe environment for experimentation, Faster analytics processing, Supports multiple analytical tools, Helps in collaboration among teams

ETLT Process - Extract → Transform → Load → Transform

It is a modern data integration process used in Big Data systems.

Data Sources



Difference between ETL and ETLT

ETL	ETLT
Transform before loading	Transform after loading
Used in traditional systems	Used in Big Data systems
Less scalable	Highly scalable
Limited processing	Supports advanced analytics

Conclusion

The Data Preparation phase converts raw data into clean and structured data for analysis. Analytics Sandbox provides a safe environment for experimentation, while the ETLT process helps efficiently process and transform large-scale Big Data for analytics.

c) List Common Tools Used for Model Building Phase of Data Analytics

Common Tools Used in Model Building Phase

Tool	Description
Python	Popular programming language used for machine learning and analytics
R Programming	Used for statistical analysis and visualization
Apache Spark MLlib	Machine learning library for big data processing
SAS	Advanced analytics and statistical software
MATLAB	Used for mathematical modeling and simulation
RapidMiner	Data mining and predictive analytics tool
Weka	Open-source machine learning tool
TensorFlow	Deep learning and neural network framework
Scikit-learn	Python library for machine learning algorithms
KNIME	Open-source analytics platform

Tool	Description
Hadoop Mahout	Scalable machine learning framework for Hadoop
Excel	Basic statistical analysis and visualization

Functions of Model Building Tools - Data analysis, Machine learning, Prediction, Classification, Clustering, Visualization

Conclusion

Various tools such as Python, R, Spark MLlib, TensorFlow, and Weka are used in the model building phase to create accurate analytical and predictive models.

Q1b) Difference Between Business Intelligence (BI) and Data Science

Data Science is an interdisciplinary field that uses Statistics, Machine Learning, Data Analytics, Programming to extract useful insights and knowledge from data. It helps organizations make intelligent and data-driven decisions.

Applications of Data Science - Fraud detection, Recommendation systems, Weather forecasting, Healthcare analytics, Sales prediction

Business Intelligence (BI)	Data Science
Focuses on past and present data	Focuses on future predictions
Uses reports and dashboards	Uses machine learning and statistical models
Answers "What happened?"	Answers "What will happen?"
Mainly descriptive analytics	Predictive and prescriptive analytics
Uses structured data	Uses structured and unstructured data
Less complex	More complex and analytical
Used for business reporting	Used for prediction and intelligent decision-making
Example- BI: Monthly sales report generation.	Example- Data Science: Predicting future sales using machine learning.

Q2a) What are the Three Characteristics of Big Data? What are the Main Considerations in Processing Big Data?

Three Characteristics of Big Data (3Vs)

Big Data is mainly characterized by three important properties called the 3Vs.

1. Volume

Volume refers to the huge amount of data generated every second.

Examples - Social media posts, Online transactions, Sensor data

Challenge - Storing and managing large datasets.

2. Velocity

Velocity refers to the speed at which data is generated and processed.

Examples - Live stock market updates, Online streaming, GPS tracking

Challenge - Real-time data processing.

3. Variety

Variety refers to different types and formats of data.

Types

- Structured data
- Semi-structured data
- Unstructured data

Examples - Text, Images, Videos, Emails

Main Considerations in Processing Big Data

1. Data Storage

Efficient storage systems are required for handling massive data volumes.

Example - Hadoop Distributed File System (HDFS)

2. Data Processing Speed

Data should be processed quickly, especially real-time data.

Example - Apache Spark

3. Scalability

System should support increasing data size and users.

4. Data Security

Protect sensitive data from unauthorized access.

Methods - Encryption, Authentication

5. Data Quality

Ensure data is Accurate, Complete, Consistent

6. Fault Tolerance

System should continue working even if some components fail.

7. Data Integration

Combine data from multiple sources efficiently.

Conclusion

The three main characteristics of Big Data are Volume, Velocity, and Variety. Efficient processing of Big Data requires proper storage, scalability, security, high-speed processing, and quality management.

Q2b) Write short note on -

i) ETL (Extract, Transform, Load)

ETL is a process used in Data Warehousing and Data Analytics to collect data from different sources, convert it into a suitable format, and load it into a database or data warehouse.

Steps in ETL Process

- 1. Extract** - Data is collected from different sources such as Databases, Websites, ERP systems, Social media
- 2. Transform** - Collected data is cleaned and converted into proper format.
- 3. Load** - Transformed data is loaded into Data warehouse, Hadoop system, Analytics database

Advantages of ETL

- Improves data quality
- Combines data from multiple sources
- Supports analytics and reporting
- Enables faster decision-making

Conclusion

ETL is an important process used to prepare and integrate data for business intelligence and analytics applications.

iii) Model Selection for Data Analytics

Model Selection is the process of choosing the most suitable analytical or machine learning model for solving a specific business problem.

The selected model should provide better accuracy and performance.

Objectives of Model Selection

- Improve prediction accuracy
- Reduce errors
- Select efficient algorithm
- Achieve better business results

Factors Affecting Model Selection

1. Nature of Problem

Problem Type Suitable Model

Prediction Linear Regression

Classification Decision Tree

Clustering K-Means

2. Type of Data

- Structured data
- Unstructured data
- Numerical data
- Text data

3. Size of Dataset

Large datasets require scalable models.

4. Accuracy Requirement

Choose model with high accuracy and low error.

Evaluation Metrics Used - Accuracy, Precision, Recall, RMSE, R-Squared

Conclusion

Model selection is an important step in Data Analytics because selecting the correct model improves prediction accuracy and business decision-making.

Q1a) What is Driving Data Deluge? Explain with One Example.

Data Deluge

Data Deluge refers to the enormous and continuous generation of data from various digital sources at very high speed.

The rapid growth of internet usage, smart devices, and online services has resulted in massive amounts of structured and unstructured data.

Factors Driving Data Deluge

1. Social Media Growth

Platforms like Facebook, Instagram, YouTube generate huge volumes of Posts, Images, Videos, Comments every second.

2. Internet and Mobile Usage

Millions of users access Websites, Mobile apps, Online services which continuously generate data.

3. IoT and Sensor Devices

Smart devices and sensors generate real-time data.

Examples - Smart watches, Smart meters, GPS devices

4. E-Commerce and Online Transactions

Online shopping and digital payments produce massive transactional data.

5. Multimedia Content

High-resolution Images, Audio, Videos contribute significantly to data growth.

Example of Data Deluge - Social Media Platform

Every minute on YouTube:

- Thousands of videos are uploaded.
- Millions of users watch, comment, and share content.

This creates massive amounts of data every second, leading to data deluge.

Conclusion

Data deluge is driven by social media, IoT devices, internet usage, online transactions, and multimedia content. Managing this huge volume of data requires Big Data technologies and analytics tools.